



Working Paper

Getting Smart about Africa's mineral wealth

How Digital Public Infrastructure can
revolutionise mineral fiscal regimes in
West Africa

Bright Simons

February 2026

Abstract

This paper argues that West African states can close the gap between mining policy and practice by adopting Digital Public Infrastructure (DPI). Converting static mining codes into simplified, computable Rules as Code, implementing machine-readable fiscal regimes, and applying Ricardian-style smart-contract protocols can shift governance from passive rent collection to dynamic, data-driven resource management.

Using Ghana's Ewoyaa Lithium Project and its transition from a fixed to a sliding-scale royalty as a case study, we show why this digital shift is increasingly necessary. We propose a low-stakes pilot in the feldspar value chain to build institutional capability and support future harmonisation across WAEMU and the AfCFTA.

Readers are encouraged to reproduce material for their own publications, as long as they are not being sold commercially. ODI Global requests due acknowledgement and a copy of the publication. For online use, we ask readers to link to the original resource on the ODI Global website. The views presented in this paper are those of the author(s) and do not necessarily represent the views of ODI Global or our partners.

© ODI Global 2026

ODI Global
4 Millbank
London SW1P 3JA
United Kingdom
www.odi.org

This work is licensed under CC BY-NC-ND 4.0.

How to cite: Simon, B. (2026) Getting Smart about Africa's mineral wealth: how Digital Public Infrastructure can revolutionise mineral fiscal regimes in West Africa. ODI Global Working paper. London: ODI Global (www.odi.org/en/publications/getting-smart-about-africas-mineral-wealth-how-digital-public-infrastructure-can-revolutionise-mineral-fiscal-regimes-in-west-africa).

Contents

1 Introduction	4
2 Katanomics invades the Atlantic lithium space	6
2.1 Ewoyaa and its discontents	6
2.2 The shift to sliding scale royalties: a pragmatic compromise?	7
2.3 The role of critical policy audiences in mitigating policy-makers' deafness to sound critiques.....	8
3 Fiscal regime analysis: asymmetry, sensitivity and 'Shadow Royalties' ...	9
3.1 Scenario analysis: 5% vs. 10% royalty baselines	9
3.2 The 'Shadow Royalty Concession': the Growth and Sustainability Levy (GSL)	10
4 Structural flaws: the 'Cliff Effect' and the case for smoothing algorithms	11
4.1 Analysing the Cliff Edge	11
4.2 Smoothing via interpolation is a simple, smart compromise.....	12
5 Operationalising policy: the Digital Gazette and 'SmartFiscal Rules' enforcement.....	13
5.1 Encoding legislation in a Digital Gazette	13
5.2 Ricardian Smart Contracts for automated metering	14
5.3 Data sovereignty and counter-extraction.....	14
6 Regional harmonisation and the low-stakes pilot: eyes on Feldspar	15
6.1 Dissecting harmonisation failure in WAEMU/ECOWAS	15
6.2 The Ceramics (Feldspar) Pilot: a Trojan Horse for reform	15
7 Conclusion: scaling 'Resource Nationalism' to 'Smart Sovereignty'	18
References.....	20

1 Introduction

As concerns over climate change force a shift to technologies like batteries, a new class of minerals – such as lithium, cobalt and rare earth elements – sometimes called ‘critical minerals’ have taken centre stage in the age-old struggle over the Earth’s natural wealth.

For West Africa, a region historically tied to ‘older’ minerals like gold and bauxite, how this transition unfolds will determine whether it can turn its minerals into a source of economic progress. This is an all too pressing imperative considering that the sub-region ranks last but one, just above Central Africa, on the list of the world’s poorest geopolitical units as defined by the UN Geoscheme.

The narrative of African mineral wealth is often framed through a lens of abundance. A pernicious myth has taken root (call it the ‘myth of 30%’) that suggests the continent holds a third of the world’s resources and needs only to wait for global demand to come knocking.

This abundance mindset obscures a more precarious reality: only a few countries are rich in very specific categories of minerals. The majority of countries on the continent are actually so poor in industrial minerals that, should industrial production pick up, mineral supply challenges would pose an immediate economic security issue. The intriguing thing is that Europe faced the exact opposite situation when it was industrialising in the nineteenth century: it was a global leader in industrial minerals like iron and coal. We detail this strand of analysis in a companion paper.

Africa should be in an unholy rush to find more minerals, and to optimise its revenues from its current mineral stock with far greater anxiety and determination. The *abundance myth* does not create this state of mind.

in competing jurisdictions like Australia or Brazil. Furthermore, the fiscal instruments used to capture value from these resources – such as static royalty rates, negotiated concession agreements and manual tax Both quests – finding more minerals and optimising revenues from the current mineral base – are data-intensive. But the continent is ‘data-poor’ in an era where geological intelligence is the primary currency of exploration, and its fiscal regimes are structurally ill-equipped to manage the volatility and complexity of modern mineral markets.

The challenge facing nations like Ghana, Mali and Burkina Faso is not merely one of extraction but of 'Sovereign Intelligence'. In the absence of high-fidelity, sovereign-managed geological data, African governments effectively sell lottery tickets called 'exploration licences' without knowing the odds of winning, while foreign entities utilise advanced Artificial Intelligence (AI) and 'mirror geology' techniques to extract value from African data to de-risk investments assessments – suffer from what we term 'Katanomics', a structural fracture between political ambition and policy execution.

This paper argues that bridging this gap requires the deployment of Digital Public Infrastructure (DPI). By transforming static mining codes into a simpler, more manageable version of what elsewhere is known as Rules as Code (RaC), implementing *computable fiscal regimes*, and utilising protocols similar to *Ricardian Smart Contracts* (Vavilis et al., 2024) for enforcement, West African states can shift from passive rent-seeking to dynamic, data-driven resource management.

We utilise the case study of Ghana's Ewoyaa Lithium Project, specifically the shift from a fixed to a sliding-scale royalty, to demonstrate the necessity of this digital transition. We further propose a low-stakes pilot using the feldspar value chain. This learning-by-doing approach aims to build the policy stamina required for regional harmonisation under the West African Economic and Monetary Union (WAEMU) and the African Continental Free Trade Area (AfCFTA).

2 Katanomics invades the Atlantic lithium space

2.1 Ewoyaa and its discontents

The discovery of the Ewoyaa pegmatite deposit in Ghana's Central Region by Atlantic Lithium (formerly IronRidge Resources) marked a pivotal moment in the country's mining history. Unlike the established gold sector, which operates under a century of accumulated legal precedent and infrastructure, the lithium sector is a 'greenfield' regulatory space. The Ewoyaa project, poised to be Ghana's first lithium-producing mine, became the testing ground for the country's Green Minerals Policy, a framework intended to break the colonial cycle of raw material export and enforce domestic value addition.

However, the political narrative of 'historic value capture' collided with the operational reality of the project's developer. Atlantic Lithium, a small Australian mining start-up, is far from a diversified mining major with a fortress balance sheet; it is a junior explorer navigating the treacherous terrain between discovery and production. As of September 2025, the company reported cash reserves of only A\$4.1 million, a critically low figure for a company approaching a Final Investment Decision (FID) estimated at \$185 million. This financial fragility, coupled with reliance on funding from offtake partners like Piedmont Lithium, has led to the realisation that the state was negotiating for something more than just a share of profits. The state's negotiators appeared to be negotiating to safeguard the very solvency of the operator.

It is a common enough dynamic, and one that I consider a classic manifestation of Katanomics. Defined as the condition where the chains linking politics (promises), policy (design), law (binding rules) and constitution (norms) are fractured, Katanomics explains why democratic pressure often fails to yield developmental results. In the case of Ewoyaa, the political imperative was to maximise state revenue and enforce 'resource nationalism'. The policy reality, however, required a nuanced understanding of the project's Internal Rate of Return (IRR) and the global lithium price crash of 2023–2024. The disconnect between the political rhetoric of 'windfall profits' and the technical reality of a marginal project

created a state of 'political dysphonia',¹ where intense public debate, led by civil society organisations (CSOs) like IMANI, generated noise but struggled to cohere into a viable policy alternative.

2.2 The shift to sliding scale royalties: a pragmatic compromise?

In October 2023, the Ghanaian government signed an initial mining lease with Atlantic Lithium featuring a 10% royalty and a 13% free carried interest. The headline rate was double the standard 5% gold royalty, a fact the government touted as evidence of its commitment to securing a 'better deal' for critical minerals.

However, using a fixed-rate regime was fundamentally flawed because it failed to account for the extreme volatility of lithium prices, which had swung from under \$500/ton to over \$6,000/ton in a mere two years. Little regard seems to have been paid to the complex chemistry-economics of transition minerals like lithium, and the geopolitical context in which even traditionally liberal mining jurisdictions like Chile were nationalising such resources to preserve strategic options.

Following the withdrawal of the lease in 2024 due to CSO pressure, the government introduced the Minerals and Mining (Royalty) Regulations 2025. The new instrument replaced the fixed 10% rate with a sliding-scale mechanism, pegged to the price of Spodumene Concentrate (SC6). The bands were set as follows:

- 5% royalty for prices < US\$1,500/ton.
- 7% royalty for prices > US\$1,500 < US\$2,500.
- 10% royalty for prices > US\$2,501 and < \$3,000.
- 12% royalty for prices > US\$3,000.

The proposed adjustment is an implicit admission of the fragile financial circumstances of Atlantic Lithium. With market prices at the time of the scale's proposal hovering between \$800 and \$1,200 per ton, the effective royalty rate dropped to 5%, a 50% reduction from the original deal. While this concession was interpreted by some analysts as necessary to ensure the project's bankability, it highlighted the weakness of a system that relies on fixed legislative provisions rather than dynamic, algorithmic regulation. The confusion surrounding the flip-flopping motives of the government also did little to bolster confidence. Policy stamina crumbled in the face of market volatility and operator fragility, a symptom of a governance system that lacks the critical policy audiences capable of sense-checking based on rigorous debate and analysis.

¹ 'Dysphonia' is borrowed from the medical literature. It refers to a voice disorder in which a person's speech is distorted due to issues with the vocal cords.

2.3 The role of critical policy audiences in mitigating policy-makers' deafness to sound critiques

The concept of 'critical policy audiences' is central to escaping the Katanomic trap. These are detail-oriented constituencies – ranging from technocrats and specialised journalists to data-literate CSOs and citizens in the analytical professions – capable of insulating policy execution from the volatility of political theatre.

In the Ewoyaa saga, the absence of coherent dialogue among a critical policy audience led to a binary and polarised debate: either 'resource nationalism' (keep the 10%) or 'investor friendliness' (drop to 5%). A more sophisticated audience might have argued for a *Computable Contract* approach from the outset. By that we mean one where the royalty rate is a continuous function of operating margin rather than a crude price-pegged step function. In Ghana's case, even the basic minimum of an updated feasibility study to gauge the fiscal soundness of the proposed royalty revision was not forthcoming.

The absence of clear policy signals resulted in a delay in ratification and a loss of investor confidence. The eventual compromise, seemingly pragmatic, introduced new structural risks, specifically an unconscionably low baseline royalty, the 'Cliff Effect' in the proposed sliding scale, and ambiguity regarding price determination (e.g. FOB vs. CIF), which traditional *analogue governance* is ill-equipped to police.

The key lesson from the Ewoyaa saga is the need to shift towards DPI, thereby transforming the 'passive' state into an 'active', data-driven sovereign.

3 Fiscal regime analysis: asymmetry, sensitivity and 'Shadow Royalties'

To understand the true impact of the shift from a fixed 10% royalty to a 5% baseline sliding scale, one must move beyond headline numbers and perform a rigorous sensitivity analysis of the 'Effective Government Take'. Such modelling reveals that, while the state made a concession on the royalty line, it is management of the broader fiscal architecture that ensures a solid revenue floor for protecting the state in downside scenarios.

3.1 Scenario analysis: 5% vs. 10% royalty baselines

We model the fiscal split for one ton of SC6 at a benchmark price of \$1,200/ton, utilising cost assumptions from the Definitive Feasibility Study (DFS) which places the All-In Sustaining Cost (AISC) at US\$610/ton.

Table 1 Comparative fiscal analysis (per ton basis)

Fiscal instrument	Scenario A: Old deal (10% fixed)	Scenario B: New deal (5% base)	Variance (state perspective)
Gross Revenue	\$1,200.00	\$1,200.00	-
(-) AISC	\$610.00	\$610.00	-
Royalty	\$120.00 (10%)	\$60.00 (5%)	-\$60.00 (Direct Loss)
GSL (1% Non-deductible)	\$12.00	\$12.00	\$0.00
CDF (1% Deductible)	\$12.00	\$12.00	\$0.00
Taxable Income Base	\$446.00	\$506.00	+\$60.00

<i>Calculation</i>	<i>(Rev - AISC - Roy - CDF)</i>	<i>(Rev - AISC - Roy - CDF)</i>	<i>Lower royalty increases tax base</i>
Corporate Income Tax (35%)	\$156.10	\$177.10	+\$21.00 (Tax Shield Effect)
Net Profit (Post-Tax)	\$277.90	\$316.90	+\$39.00
GoG Dividend (13% Free Carry)	\$36.13	\$41.20	+\$5.07
Total State Take	\$324.23	\$290.30	-\$33.93
Investor Net Cash Flow	\$241.77	\$275.70	+\$33.93

Source: Author's calculations

Key insight: the asymmetry of concession

The analysis reveals a counter-intuitive finding: a 100% doubling of the baseline royalty to protect the state's downside in extreme price crash scenarios only suppresses the net profit level of the private investor (Atlantic Lithium) by a little more than 10%. The downside to the investor's bottom line is limited while the sovereign fiscal bulwark in a bear market scenario is markedly improved, as we shall demonstrate shortly.

3.2 The 'Shadow Royalty Concession': the Growth and Sustainability Levy (GSL)

A critical component of Ghana's fiscal regime that often escapes scrutiny is the Growth and Sustainability Levy (GSL). Unlike the royalty, which is a payment for the mineral asset, the GSL is a fiscal stabilisation tool. For the lithium sector, it is set at 1% of gross production, far lower than for legacy gold miners, who face a 3% rate. The effect of the concessions to Atlantic Lithium is that the baseline effective royalty rate for a mature mining sector, that of gold, is now higher than for a sizzling hot sector with massive upside prospects, low production costs and considerable equity participation by the state (in percentage of development cost terms).

The 'sting' of the GSL lies in its non-deductibility. It is not treated as an operating expense for CIT calculations. This means it functions as a 'Shadow Royalty' or a 'Super-Revenue' tax. For the legacy gold sector, even in a 'Bear Case' scenario where gold prices drop below \$1,900/ton and margins literally disappear, the state continues to extract revenue at a whopping effective royalty rate above 8%.

4 Structural flaws: the 'Cliff Effect' and the case for smoothing algorithms

Focusing strictly on lithium, we see an even more intriguing flaw. While the sliding scale is an improvement over a fixed rate, the implementation chosen by Ghana, i.e. a 'step' or 'bracket' system, introduces a severe structural flaw known as the 'Cliff Effect'. This occurs when the royalty rate jumps instantaneously at a specific price threshold, creating massive marginal tax rates that distort economic behaviour. Similar concerns and possible remedies were explored by IMF analysts nearly three decades ago (Baunsgaard, 2001).

4.1 Analysing the Cliff Edge

The 2025 Regulations specify that the royalty jumps from 7% to 10% when the price exceeds \$2,500/ton. Let us analyse the economic incentives for a mining company operating near this threshold.

Scenario A: Honest pricing at \$2,510/ton

- The price is above \$2,500, so the 10% rate applies to the *entire* gross revenue (rather than just the incremental amount).
- Gross revenue: \$2,510.00
- Royalty payment: $\$2,510 \times 0.10 = \251.00
- Net revenue (pre-tax): $\$2,510 - \$251 = \$2,259.00$

Scenario B: 'Gaming the price' at \$2,500/ton

- The company artificially suppresses the invoice price (e.g. by inflating transport costs to lower the FOB price) to exactly \$2,500.
- Gross revenue: \$2,500.00
- Royalty rate: 7% (Band 2 applies).
- Royalty payment: $\$2,500 \times 0.07 = \175.00
- Net revenue (pre-tax): $\$2,500 - \$175 = \$2,325.00$

A perverse incentive

By selling the lithium for \$10 less, the company actually makes \$66 more in net revenue (\$2,325 - \$2,259). The result is a 'dead zone' above the threshold where it is economically irrational to sell at the market price. This \$66 arbitrage creates a powerful incentive for transfer pricing manipulation, under-invoicing or 'gold-plating' logistics costs to keep the realised FOB price just below the cliff edge.

4.2 Smoothing via interpolation is a simple, smart compromise

To eliminate this perverse incentive, the fiscal regime must transition from discrete steps to a continuous, smoothed curve. We propose the adoption of a Smooth Interpolation Algorithm, which gradually increases the effective royalty rate as the price moves through the band. The simplest one will be linear.

Mathematical logic

Instead of a jump, the royalty rate R is calculated as a function of the price P using the following formula for the band between \$1,500 and \$2,500 (moving from 5% to 7%):

$$R = 0.05 + \left(\frac{P - 1500}{1000} \times (0.07 - 0.05) \right).$$

Re-evaluating the \$2,510 scenario with smoothing:

- At \$2,500, the rate is exactly 7%. Royalty = \$175.
- At \$2,510, the rate is interpolated between 7% and 10%.
 - Band width = \$500 (\$3,000 - \$2,500). Rate Delta = 3% (10% - 7%).

$$R = 0.07 + \left(\frac{2510 - 2500}{500} \times 0.03 \right) = 0.07 + 0.0006 = 7.06\%.$$

- Royalty payment: \$2,510 x 0.0706 = ~\$177.20
- Net revenue: \$2,510 - \$177.20 = \$2,332.80

Outcome

Under the smoothed model, earning more revenue (\$2,510) results in higher net profit (\$2,332.80) compared to the 'gaming' scenario. The incentive to manipulate prices is mathematically minimised to an insignificant level. This approach ensures that the state's share increases progressively without breaking the economic logic of the operation.

5 Operationalising policy: the Digital Gazette and 'SmartFiscal Rules' enforcement

The existence of a superior mathematical model is insufficient if the administrative machinery cannot implement it. In the 'Katanomic' state, policies often fail because they rely on poorly scrutinised and supervised processes, which are prone to error and manipulation. To operationalise the smoothed royalty regime, we propose the implementation of DPI centred on a 'Digital Gazette' and a 'SmartFiscal' protocol.

5.1 Encoding legislation in a Digital Gazette

Currently, mining fiscal terms are promulgated in government Gazettes as static PDF documents or printed text. This format contributes to policy atrophy because the rules are ambiguous, hard to parse for automated systems and disconnected from the point of transaction. A Digital Gazette transforms legislation from static text into computable referents (Mohun and Roberts, 2020).

Proposed architecture

- 1 Repository of Truth: The Minerals Commission publishes the fiscal logic (e.g. the linear interpolation formula) but does so in both legal prose and a supplementary standardised code block (e.g. JSON, Python or YAML) in a public, version-controlled repository (similar to GitHub for Law) (Rapson et al., 2025).
- 2 Versioning and immutability: Any change to the fiscal regime is committed as a new version, thereby creating an immutable audit trail of exactly which formula was in force at any given timestamp, eliminating disputes over 'interpretation' (Organisation for Economic Co-operation and Development Public Sector Experimentation Team, 2024).
- 3 Public auditability: Critical policy audiences (CSOs, journalists and investors, among others) can run simulations against this code to verify its impact. They can use tools like the 'Cliff Analyzer' (Marques Junior, 2024) to test for edge cases before the law is ratified, effectively crowd-sourcing policy quality assurance.

5.2 Ricardian Smart Contracts for automated metering

To enforce the rules held in the Digital Gazette, the SmartFiscal system could be made to interface with the physical movement of goods via Ricardian Smart Contracts. By chaining supply chain events to the analyser module, price index fluctuations that can alter realised gross proceeds can be tracked in real-time to establish smoothed-averages of total realised pricing for shipped commodities. A Ricardian Contract is dual-natured: it is a legally binding agreement readable by humans and a machine-parsable script executable by software (Brickken Solutions S.L., 2025).

A smart-contract approach is not an absolute requirement for the main process to run. What is important in this specific scenario is to have a user interface that outputs the exact tax amount along a smoothed curve in any operational and accounting medium.

5.3 Data sovereignty and counter-extraction

One could go further in the use of digital tools to supercharge natural resource financial outturn by moving further upstream to the exploration level. Such a DPI framework would, however, require a new component: the National Data Cloud or Standardized Exploration Data Room, which we explain in greater detail in the companion paper co-published with this policy brief.

Currently, foreign entities have a massive incentive to use AI to 'mine' Africa's geological data, identifying patterns ('mirror geologies') to de-risk investments in other jurisdictions where the base mining data quality makes such models more valuable.

By mandating that all exploration data be uploaded to a sovereign cloud in machine-readable formats (LAS, CSV) with QA/QC metadata, the state asserts ownership over its 'Geological IP'. The ensuing data becomes a strategic asset that can be licenced to AI companies or used to lower the barrier to entry for new investors, shifting the nation from a passive owner to an active manager (Ortega Girones et al., 2009) of its resources. This is a hugely critical need, given how many African countries remain dependent on low-resolution cadastral maps, some from the colonial period (Lasselin and Lequeux, 2009).

Furthermore, pure exploration companies can now be incentivised to map Africa's mining belt at a much higher resolution and to upload AI-enriched data to the National Cloud under a special digital licencing regime utilising the same SmartFiscal logic, but this time for B2B royalties.

A chain-of-use and chain-of-benefit model provides a mechanism for windfall benefits to accrue to the original exploration data providers when discoveries are made by companies that have benefited from the geological IP trail.

6 Regional harmonisation and the low-stakes pilot: eyes on Feldspar

6.1 Dissecting harmonisation failure in WAEMU/ECOWAS

The challenge of fiscal governance is not contained within national borders. The West African Economic and Monetary Union (WAEMU) and ECOWAS have spent decades attempting to harmonise tax and mining codes to prevent a 'race to the bottom' where countries underbid each other to attract investment. Despite directives like C/DIR.3/05/09 (discussed, for example, on page 16 of this document), harmonisation has been patchy in practice (Traore, 2024). Member states retain broad discretion, granting extensive exemptions that undermine the regional framework (Mansour and Rota-Graziosi, 2013).

The root cause is political economy: asking a sovereign state to cede control over the fiscal terms of its primary revenue generator (whether gold, lithium or petroleum) to a regional body is a heavy political lift. The stakes are too high, and the policy stamina too low. To break this deadlock, the region needs a 'sandbox': a low-stakes environment to test DPI and harmonisation protocols without threatening national budgets.²

6.2 The Ceramics (Feldspar) Pilot: a Trojan Horse for reform

We propose Feldspar as the ideal candidate for this regional sandboxing venture.

- *Resource attributes:* Feldspar is a major byproduct of lithium projects in the region, for example in the Ewoyaa Lithium Project. Atlantic Lithium's updated Mineral Resource Estimate reports 36.8Mt at 41.9% feldspar. Currently, this material is considered waste or a secondary credit.

² Sandboxes have become a mainstay of the policy innovation space worldwide. See Allen, H.J. (2019); Buckley et al. (2020); Jenik and Duff (2020); Ranchordás and Vinci (2024). There are also fascinating parallels, such as Sustainability Directory (n.d.).

- *Market dynamics:* West Africa has a burgeoning ceramics industry (producing, among other things, tiles and sanitary ware) in Ghana and Nigeria, which is currently hindered by the lack of processed raw materials. The region imports feldspar despite having vast deposits.
- *Low stakes:* Unlike gold or lithium, feldspar is not a 'critical' security mineral for the West, nor is it a primary fiscal lifeline for the state. This lowers the political temperature, creating an environment where technocrats and policy-shapers can experiment with new governance models.

Initiating the logic

- 1 Cross-border digital rails: Establish a digital trade corridor for feldspar between Ghana (producer) and Nigeria/Togo (consumers) using the AfCFTA digital protocol.
- 2 Harmonised coding: Implement the Digital Gazette and SmartFiscal protocol standards for feldspar royalties. Because the revenue is negligible compared to gold, resistance from vested interests will be reduced.
- 3 Testing the pipes: Use the pilot to debug the 'Rules as Code' infrastructure, such as fixing the cliff effects, refining the Oracles and testing the interoperability of National Data Clouds. Once the plumbing is in peak condition, consider scaling the architecture to higher-stakes sectors (in ascending order of sensitivity: limestone, gypsum, phosphate, lithium and gold). Naturally, the digital mineral royalties chain-of-benefits mechanism would increase in value as the stakes increase towards higher-value minerals like gold and lithium.
- 4 Code the harmony: Explore emergent techniques for calibrating national rules to configure incentives for investors to prioritise connectors in regional value chains. Once stable, build an optimisation algorithm to estimate the ideal rates to be charged in each jurisdiction for specific fiscal items such that, put together, the region as a whole becomes more competitive.

The strategic ambition should be to get West Africa to build the institutional muscle and policy stamina required for sovereign control, starting with the humble ceramic tile before advancing to the electric vehicle battery.

It is important to emphasise that the SmartFiscal protocol approach, including the digital minerals royalty model, stops short of some of the overly techno-optimistic directions being pursued by some in the Rules-as-Code movement. Self-executing triggers are not central to the success of the SmartFiscal and related protocols, thus escaping some of the deterministic excesses criticised in the literature.³ The

³ See a small subset of the litany of critiques that has mounted against naïve Rules-as-Code automation stampedes: Brownsword (2019); Coglianese and Lehr (2019); Hildebrandt (2018); Lessig (1999).

emphasis here is on surgical inserts within legal regimes, such as fiscal calibrators, that mimic obsolete technology (Gilson et al., 2010), and how to upgrade them without corroding the legal constructs in which they are embedded.

7 Conclusion: scaling 'Resource Nationalism' to 'Smart Sovereignty'

The transition from the 'Old Deal' (fixed royalties) to the 'New Deal' (sliding scale) at Ewoyaa is a microcosm of Africa's broader struggle to master its resource endowments. While the December 2025 Regulations represent a pragmatic compromise towards investor solvency and state value capture, they remain vulnerable to the structural flaws of analogue governance: cliff effects, transfer pricing opacity and Katanomic policy drift.

As we have said elsewhere, DPI represents a very exciting path forward. Ghana and its West African peers can transcend these limitations with the right kind of policy smarts. The adoption of smoothing algorithms, for instance, eliminates perverse incentives and embeds far more flexible fiscal elasticities into the full suite of fiscal regime technologies, well beyond royalties. Digital Gazettes and Rules as Code transform passive legislation into active, enforceable and responsive logic.

At the peak of a jurisdiction's digital maturity, it can even explore Ricardian Smart Contracts to ensure that fiscal terms are executed as agreed, without the friction of manipulatable bureaucratic discretion. A state may go as far as directly embedding such logic into electronic fiscal devices being rolled out incrementally in various countries in the region (Bassongui and Houngbédji, 2021) and/or the electronic resource platforms of more mature resource companies.⁴

We see such a prospective trend as, ultimately, representing a shift from 'Resource Nationalism', a phenomenon that has been over-politicised in Africa and has barely transcended the rhetorical level, to 'Smart Sovereignty'. We consider this a policy-first approach: smart sovereignty, carefully calibrated, can deliver real performance by granting policy-makers the stamina they need to iterate, optimise and course-correct towards increasingly superior fiscal outcomes. Vastly boosting resource value in Africa should be the endgame.

From a political economy standpoint, starting with a low-stakes feldspar sandbox, West Africa can build the digital and institutional

⁴ See, for an analytical parallel, Santoro et al. (2025) on Eswatini.

capacity required to govern its critical and transition-minerals future, ensuring that the vital substances of the energy transition fuel domestic development instead of seeping away into foreign batteries.⁵

⁵ As implicitly implied in, for example, Kosiński (2023).

References

- Allen, H. J. (2019) *Regulatory sandboxes*. *George Washington Law Review* 87: 579–612. <https://www.gwlr.org/wp-content/uploads/2019/06/87-Geo.-Wash.-L.-Rev.-579.pdf>
- Bassongui, N. and Houngbédji, H.S. (2021) Does tax digitalisation improve tax revenues collection in sub-Saharan Africa. In *Conference économique africaine*. https://aec.afdb.org/sites/default/files/papers/does_tax_digitalisation_improve_tax_revenues_collection_in_subsaharan_africa_bassongui.pdf
- Baunsgaard, T. (2001) *A primer on mineral taxation*. IMF Working Paper No. 01/139. International Monetary Fund. <https://www.imf.org/external/pubs/ft/wp/2001/wp01139.pdf>
- Brickken Solutions S.L. (2025) *Smart contracts in finance: The backbone of trustless transactions and automated compliance*. Brickken. <https://www.brickken.com/post/smart-contracts-in-finance-automation-compliance>
- Brownsword, R. (2019) *Law as a moral judgment versus law as a technical system*. *Legal Studies* 39(1): 1–24. <https://doi.org/10.1017/lst.2018.25>
- Buckley, R.P. et al. (2020) *Building FinTech ecosystems: Regulatory sandboxes, innovation hubs, and beyond*. *Washington University Journal of Law & Policy* 61: 55–115. https://openscholarship.wustl.edu/law_journal_law_policy/vol61/iss1/10/
- Coglianesi, C. and Lehr, D. (2019) *Regulating by robot: Administrative decision making in the machine-learning era*. *Georgetown Law Journal* 105: 1147–1223.
- Gilson, R. J., Sabel, C.F. and Scott, R.E. (2010) *Contracting for innovation: Vertical disintegration and interfirm collaboration*. *Columbia Law Review*, 109, 431–502. <https://charlessabel.com/papers/contracting%20for%20innovation.pdf>
- Hildebrandt, M. (2018) *Law as computation in the era of artificial legal intelligence: Speaking law to the power of statistics*. *University of Toronto Law Journal* 68(S1): 12–35. <https://doi.org/10.3138/utlj.2017-0083>
- Lessig, L. (1999) *Code and other laws of cyberspace*. Basic Books. <https://hls.harvard.edu/bibliography/code-and-other-laws-of-cyberspace-version-2-0/>

- Jeník, I. and Duff, S. (2020) *How to build a regulatory sandbox: A practical guide for policymakers*. CGAP/World Bank Group.
<https://digitalfinance.worldbank.org/sites/default/files/2022-11/How-to-Build-a-Regulatory-Sandbox-A-Practical-Guide-for-Policy-Makers.pdf>
- Kopiński, D. (2023) *African critical raw materials and the EU's economic security*. Polish Economic Institute. <https://pie.net.pl/wp-content/uploads/2023/09/Surowce-Afryki-ENG.pdf>
- Lasselin, D. and Lequeux, D. (2009) *New mapping at scales 1:200,000 of Senegal and 1:5,000 in French Polynesia: Two examples of operational production of digital topographic maps from satellite imagery*. In Proceedings of the 24th International Cartographic Conference (ICC 2009). International Cartographic Association.
- Mansour, M. and Rota-Graziosi, G. (2013) *Tax coordination, tax competition, and revenue mobilization in the West African Economic and Monetary Union*. IMF Working Paper WP/13/163. International Monetary Fund.
<https://www.imf.org/external/pubs/ft/wp/2013/wp13163.pdf>
- Marques Junior, K. (2024) From parameters to policies: sensitivity analysis and fiscal and monetary reactions. *Journal of Economic Studies* 51(7): 1514–1534.
- Mohun, J. and A. Roberts (2020) Cracking the code: Rulemaking for humans and machines. *OECD Working Papers on Public Governance* 42.
- Organisation for Economic Co-operation and Development, Public Sector Experimentation Team (2024) *Rules as Code in Canada: Summary of experiments and lessons learned* (OECD OPSI). OECD Observatory of Public Sector Innovation. <https://oecd-opsi.org/wp-content/uploads/2024/04/Rules-as-Code-in-Canada.pdf>
- Ortega Girones, E., Pugachevsky, A. and Walser, G. (2009) *Mineral rights cadastre: Promoting transparent access to mineral resources*. Extractive Industries and Development Series No. 4. World Bank.
<https://hdl.handle.net/10986/18399>
- Radin, M.J. (2012) The Fine Print, Vanishing Rights, and the Rule of Law.
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2619496
- Ranchordàs, S. and Vinci, V. (2024) *Regulatory sandboxes and innovation-friendly regulation: Between collaboration and capture*. Tilburg Law School Research Paper. <https://ssrn.com/abstract=4696442>
- Rapson, J. et al. (2025) *Rules as code for a more transparent and efficient global economy*. Policy Brief. Think7 Canada. <https://www.think7.org/publications/rules-as-code-for-a-more-transparent-and-efficient-global-economy/>
- Santoro, F., Masuku, P. and Magongo, T. (2025) Cashless Tax Systems: Voluntary vs. Mandated Digital Payments in Eswatini.

- Schlag, P. (1985) *Rules and standards*. *UCLA Law Review*.
<https://lawweb.colorado.edu/profiles/pubpdfs/schlag/schlagUCLALR.pdf>
- Sustainability Directory (n.d.) *Emotional Sandbox Environment*. *Sustainability Directory: Lifestyle*. <https://lifestyle.sustainability-directory.com/area/emotional-sandbox-environment/>
- Traore, R.J. (2024) *Tax harmonisation policy within the West African Economic and Monetary Union: Analysis and prospects*. ICTD Working Paper 200. Institute of Development Studies. <https://doi.org/10.19088/ICTD.2025.015>
- Vavilis, S., Niavis, H. and García-Mauriño, L.R. (2024) On Addressing Governance Challenges in Blockchain Networks. In *2024 6th Conference on Blockchain Research & Applications for Innovative Networks and Services (BRAINS)*. IEEE.